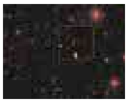
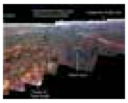


83 supermassive black-holes found
Astronomers hunting for ancient, supermassive black holes have found around 83 of them at the edge of the universe. **https://dgit.in/apr19-7**



Opportunity's last image from Mars
15 years after landing, Opportunity rover shuts down due to dust storm, sends us final panorama of Perseverance Valley. **https://dgit.in/apr19-8**

extremely effective in even more tenuous atmosphere, such as that of the Moon, and the atmosphere of Mars is just thick enough to be problematic. Retro thrusters also kick up a lot of dust and material, which could end up damaging the payload, which happened with the wind sensor on board the Curiosity rover. If airbags are used, the payloads may bounce off them and get damaged. While robotic payloads can handle the G forces generated by the air bags (between 10 to 20 Gs), humans cannot. It is also not feasible to scale the sky crane approach to the masses and sizes required for future payloads. We are at the limits of what airbags and sky cranes can handle. Humans just do not know how to land heavier payloads on Mars, and are not very good at landing even the relatively lighter ones. Though the statistics are skewed because of how popular a target Mars is, 60 percent of all attempted landings so far have failed, and the period of landing on Mars has earned the name "seven minutes of terror".

POSSIBLE SOLUTIONS

For any landing involving a payload of about 20 tons, there is a short window of time in which a lot of things need to happen. Within 90 seconds, the payload has to reduce speed from Mach 5 to Mach 1, reorient from interplanetary cruising to landing mode, reduce the remainder of the speed, identify a landing site, avoid obstacles, and execute a soft touchdown. This is known as the Supersonic Transition Problem or STP. There simply are no technologies that can decelerate large payloads fast enough... yet. Retro thrusters are the best option, but that would increase the fuel requirements, which would in turn increase the payload. The fuel required would be six times the mass of the payload itself. SpaceX and Lockheed Martin have plans for using supersonic retropropulsion on their interplanetary landers, which burn up a ton of fuel, but can safely land and take off from Mars.

There are some innovative solutions proposed though. One of them is what is called a "hypercone", an inflatable structure that acts as both a heat shield and a parachute at once. This is essentially a donut shaped structure that inflates in the upper atmosphere, with the payload housed in the center. NASA is testing out a hypercone called the Low-Density Supersonic Decelerator (LDSD) which decelerates spacecraft from upwards of Mach 3.5 to Mach 2. Once the LDSD has done its job, further reduction in speed is provided by a supersonic parachute. NASA has developed inflatable decelerators of 6 meter and 8 meter diameters, as well as a disc sail parachute made with a single sheet of fabric, and a ring sail which consists of stitched together kevlar panels. These drag devices have already been tested in the stratosphere of the Earth, and have improved the landing capacity from 1 ton to 3 tons.

In a recent paper published last month in the Journal of Spacecraft and Rockets, two researchers from the University of Illinois, Christopher G Lorenz and Zachary R Putnam came up with a novel solution. The idea is to orient the payload towards the planet, and use the supersonic speeds to generate lift, which can be used to steer the vehicle to the landing spot. The trick is to not go for a symmetrical packaging within the vehicle, shifting the center of gravity. This allows the vehicle to be heavier on one side, and fly into the atmosphere at an angle. Combining the lift generated by this angle with accurate timing for firing the retro thrusters, would allow accurately landing larger payloads. This approach actually turns out to be the best way to use the fuel on board. Putnam explains, "turns out, it is propellant-optimal to enter the atmosphere with the lift vector pointed down so the vehicle is diving. Then at just the right moment based on time or velocity, switch to lift up, so the vehicle pulls out and flies along at low altitude. This enables the vehicle to spend more time flying low where the atmospheric density is higher. This increases the drag, reducing the amount of energy that must be removed by the descent engines." The idea is to increase the time in the lower altitudes, where the spacecraft can optimally utilise as much lift as can be generated from the thin Martian atmosphere.

NASA's LDSD